

Homework 3 Due Friday March 6, 6:00pm

Problem 1 Text problem 5.5.

Problem 2 Text problem 5.10.

Problem 3 Text problem 7.3(a),(b).

Problem 4 Text problem 9.1(5). Use the following four approaches:

1. Find a control law and a corresponding Lyapunov function for the virtual control system:

$$\dot{x}_1 = -x_1 + x_1 v$$

$$\dot{x}_2 = v$$

and then use backstepping.

2. Make the preliminary choice $u = -x_1^2 - x_2 + w$, then parse the system into the form (9.23)-(9.25) in the book where x corresponds to x_1 and z corresponds to (x_2, x_3) , apply the feedback proposed in (9.30) of the text and show that the conditions of Theorem 9.2 hold.
3. Make the preliminary choice $u = -x_1^2 + w$, then parse the system into the form (9.23)-(9.25) in the book where x corresponds to (x_1, x_2) and z corresponds to x_3 , apply the feedback proposed in (9.30) of the text and explain why the origin is globally asymptotically stable using the invariance principle.
4. Based on the backstepping approach in part 1, use the control Lyapunov function approach. Verify the small control property using the second condition in Lemma 9.6 of the text.

Problem 5 Text problem 9.1(6) by choosing u to induce a cascade of an ISS system with a GAS system.

Problem 6 Text problem 9.3(a)(b) for 9.3(2).

Problem 7 Text problem 9.9(f),(g). [Part (g) amounts to writing down the control law that results from applying Theorem 9.2 in part (f).]

Problem 8 Text problem 9.13.